SOFTWARE REQUIREMENTS SPECIFICATION

For

Anti-Corruption Reporting System

Version: 1.0

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1. Introduction

1.1 Purpose

The main objective of this document is to illustrate the functional and non-functional requirements for the Anti-Corruption Reporting System. It encompasses the development of a secure web-based platform for citizens to report corruption and for administrators to manage those reports with accountability.

1.2 Document Conventions

- Entire document should be justified.
- Convention for Main title
  - Font face: Times New Roman
  - Font Style: Bold
  - Font Size: 14
- Convention for Sub title
  - Font face: Times New Roman
  - Font Style: Bold
  - Font Size: 14
- Convention for body
  - Font face: Times New Roman
  - Font Size: 12

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Supervisors(Dhrubo Barua and Md Foysal): For project validation and oversight.
- Developers: To understand technical logic and back-end requirements.
- Testers: To verify system integrity and security.
- Administrators: To understand the system’s capabilities and portal features.

1.4 Product Scope

The system serves as a centralized contact point for reporting crimes and corruption. It aims to eliminate barriers such as fear of retaliation, inefficiency in paper-based systems and lack of transparency. The scope includes secure citizen portals, admin dashboard and real-time status tracking.

1.5 References

- Project Proposal: “Anti-corruption Reporting System” (February 2026)
- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.

2. Overall Description

2.1 Product Perspective

The system is new, self-contained web application designed to replace traditional, inefficient, paper-based reporting methods. It bridges the gap between the public and law enforcement. It centralizes all corruption-related data into a single point of contact.

2.2 Product Functions

- **Secure Authentication:** Separate gateways for citizens and admins.
- **Anonymous Reporting:** Enables users to submit crimes without fear of retaliation.
- **Evidence Management:** Allows users to attach digital evidence to their complaints.
- **Real-time Tracking:** Provides a unique “Complaint ID” for users to check investigation status.
- **Admin Dashboard:** Tools for authorities to categorize, prioritize and update cases.
- **Feedback:** User rating system for administrative responsiveness.
- **Public Updates:** Administrators can share resolved case information to improve community trust.

2.3 User Classes and Characteristics

- **Citizens:** High frequency of use for reporting; requires ease of use for varied technical expertise.
- **Administrators:** High privilege levels; law enforcement personnel responsible for case updates and data management.

2.4 Operating Environment

- **Server:** Apache Web server (XAMPP)
- **Database:** MySQL database for secure data storage.
- **Platform:** Web browsers (Chrome, Firefox etc).

2.5 Design and Implementation Constraints

- **Technology:** PHP for back-end logic; HTML,CSS and JS for front-end.
- **Security:** Data transmission must be encrypted to protect whistle-blowers.
- **Budget:** Total development cost between BDT 27,000 and BDT 65,000.

2.6 User Documentation

The project will deliver documentation including a digital user manual for citizens and a technical dashboard guide for administrators.

2.7 Assumptions and Dependencies

Assumes continuous server availability(24/7) for reporting and also users have internet access and basic literacy.

3. External Interface Requirements

3.1 User Interfaces

- User Portal: Simple, clean and accessible layout for reporting forms.
- Admin Dashboard: A secure, data-rich interface for case management.

3.2 Hardware Interfaces

- The software will interface with standard web servers and client devices (PCs, laptops, smartphones) with internet connectivity.

3.3 Software Interfaces

- Database: Connection between PHP back-end and MySQL database.
- Development Tools: Use of XAMPP for hosting and Apache server management.

3.4 Communications Interfaces

The system will use HTTP/HTTPs for data transmission, ensuring encryption to protect user identities and prevent cyber attacks.

4. System Features

4.1 Secure Authentication

- Description: Separate, secure login/logout gateways for citizens and admins.
- Priority: High
- Functional Requirements: System must validate credentials against the MySQL database. Admins must have distinct privileges compared to standard users.

4.2 Anonymous Reporting

- Description: A dynamic form for users to upload evidence and describe incidents without identity requirements.
- Priority: High
- Stimulus/Response: User selects the "Anonymous" option on the reporting form. System processes the report while stripping all personal identifiers and metadata.

4.3 Complaint Management

- Description: A dynamic system for users to file complaints and provide supporting evidence.
- Priority: High
- Functional Requirements: The system shall provide a space for users to upload the evidence in the form of documents or images.

4.4 Real-time Tracking

- Description: Provides transparency by allowing users to check the status of their specific case using their unique ID.
- Priority: High
- Functional Requirements: The system shall retrieve the most recent status update from the database of a valid ID entry.

4.5 Admin Control Panel

- Description: A centralized dashboard for law enforcement to manage the influx of reports.

- Priority: High
- Functional Requirements: The system must log every status change made by an administrator to maintain an audit trail.

4.6 Feedback Loop

- Description: Priority: Medium; A dedicated section for users to rate the responsiveness of the administration.
- Functional Requirements: The system shall provide a star-rating and text-entry field for user feedback on resolved cases.

4.7 Update System

- Description: Priority: Medium; Allows administrators to post public updates on resolved cases to build community trust.
- Functional Requirements: The system shall allow administrators to draft and publish public announcements regarding general anti-corruption progress or specific resolved cases.

5. Other Non-functional Requirements

5.1 Performance Requirements

- Availability: 24/7 accessibility for reporting.
- Faster processing than paper-based systems.

5.2 Safety Requirements

- The system must prevent data loss by ensuring database integrity and regular backups within the XAMPP environment.

5.3 Security Requirements

- Data transmission must be encrypted to ensure user identities are protected.

5.4 Software Quality Attributes

- Transparency: Every report must be documented and traceable.
- Reliability: Ensure data is not “lost” like paper-based systems.

5.5 Business Rules

- Only designated law enforcement officers (Admins) can update case statuses.

6. Other Requirements

Appendix A: Glossary

- Complaint ID: A unique tracking number assigned to each report to allow users to check status without an account.
- Whistle-blower: A citizen reporting misconduct who requires anonymity to avoid retaliation.
- SQL Injection: A type of cyber-attack the system must be secured against.

Appendix B: Analysis Models

- **Data Flow Diagram (DFD):** This model illustrates how a report moves from the citizen portal, through the PHP logic, into the MySQL database and finally appears on the admin dashboard.
- **Entity-Relationship Diagram (ERD):** Maps the relationship between the users, complaints, evidence files and administrators within the database schema.
- **Use Case Diagram:** Shows the interactions between the citizens (reporting, tracking) and the admins (categorizing, updating, posting public news).

Appendix C: To Be Determined (TBD) List

- Hosting & Domain: Selection of a specific hosting provider for the final deployment phase.
- Maintenance Schedule: Finalizing the recurring update cycle for server renewal and feature additions.
- Local Legal Standards: Confirming if the digital evidence collected meets specific national regulatory requirements for judicial proceedings.